# Sprint 2 Report

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## 1. Sprint Overview

- **Your Team Name:** [**Team 3**]
- **Sprint 2 Dates:** [**10/02/2026 → 03/03/2026**]
- **Sprint Goals:**
  - *Implement web app connection for timer implementation with **Firestore persistence** and **safe validation**. data should be personal to each user and users should be able to login via email.*
  - *Create CUBE Client simulator to test and demo hardware REST API*
  - *Implement basic communication between ESP32 and FLASK SERVER using REST API*

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## 2. Sprint Board

**Sprint Board Link:** [**Trello**]
**GitHub Repository Link:** [**Github**]

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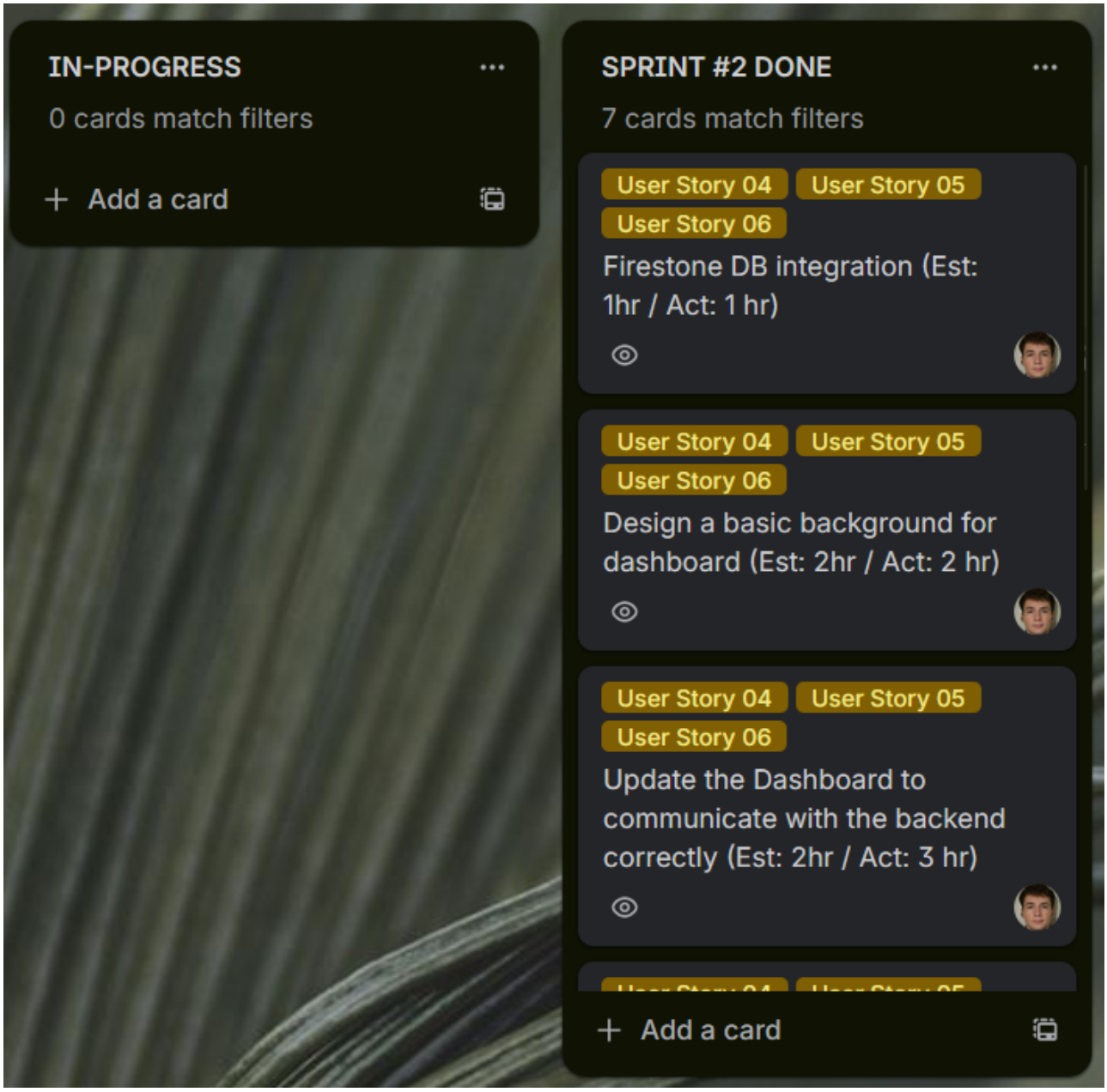
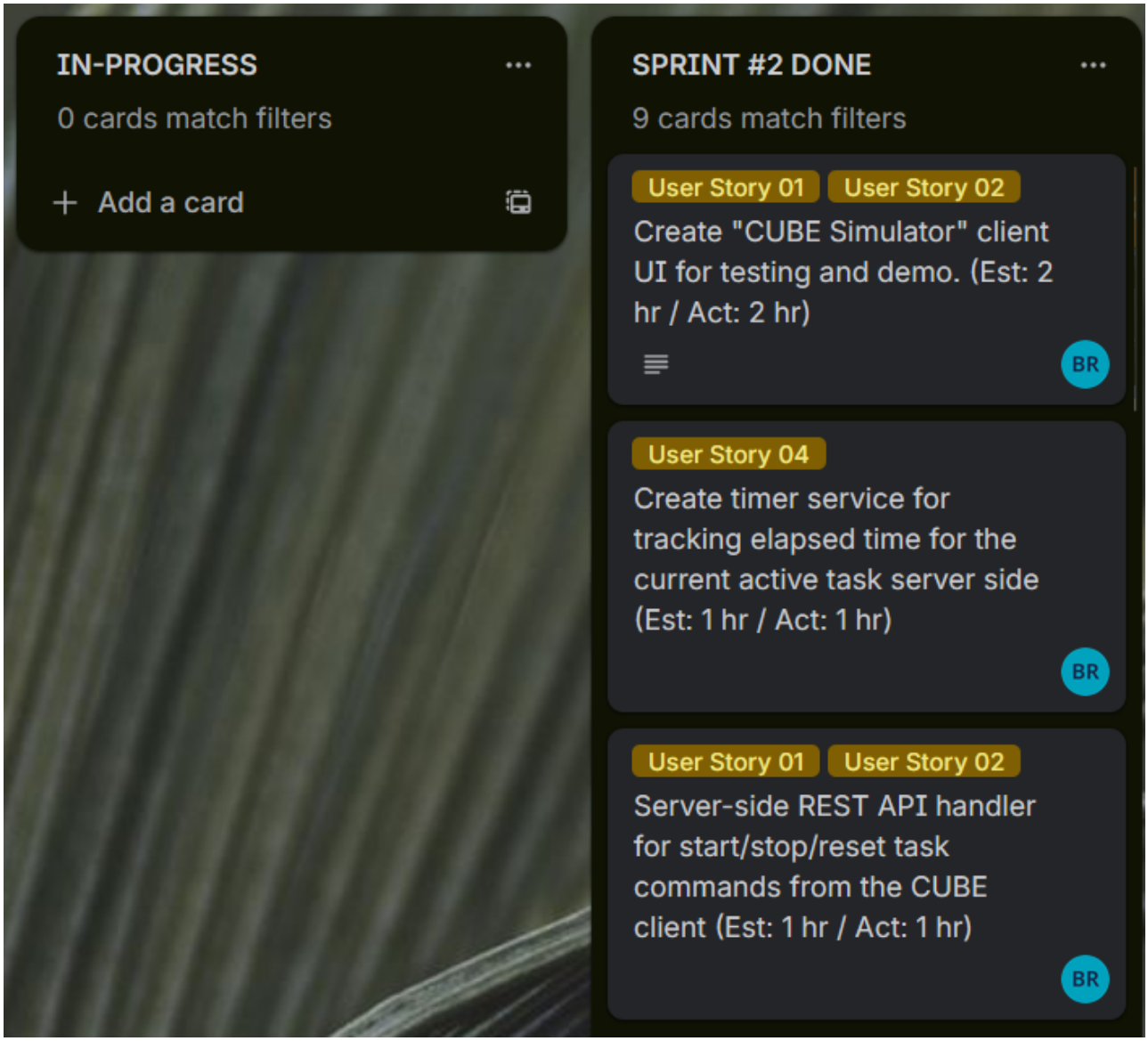
### 2.1 Sprint Board Screenshot (Filtered by Team Member)

**Please provide a screenshot of your Sprint 2 board** (e.g., Trello, GitHub Projects) **filtered by each team member**. This makes the review concrete and shows shared ownership.

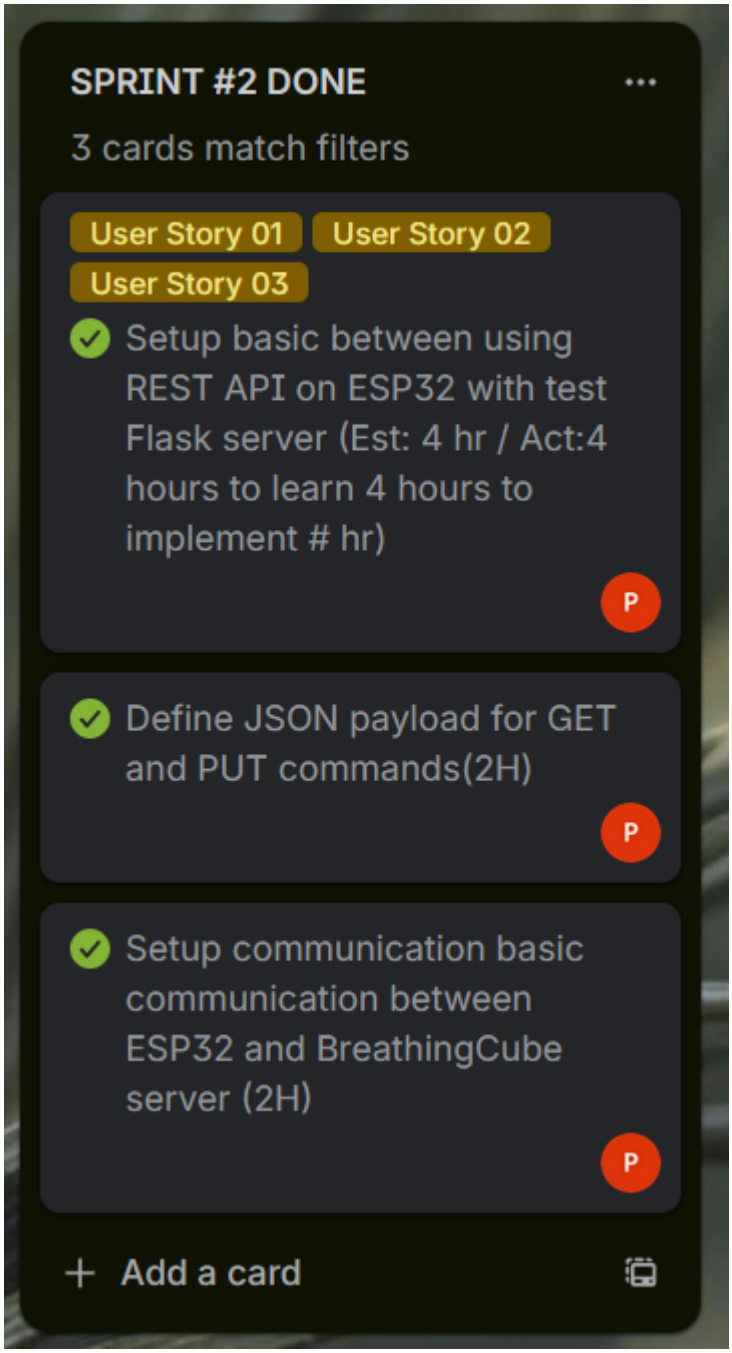
- **[Sasha Roosen-Saba] Board Screenshot:**



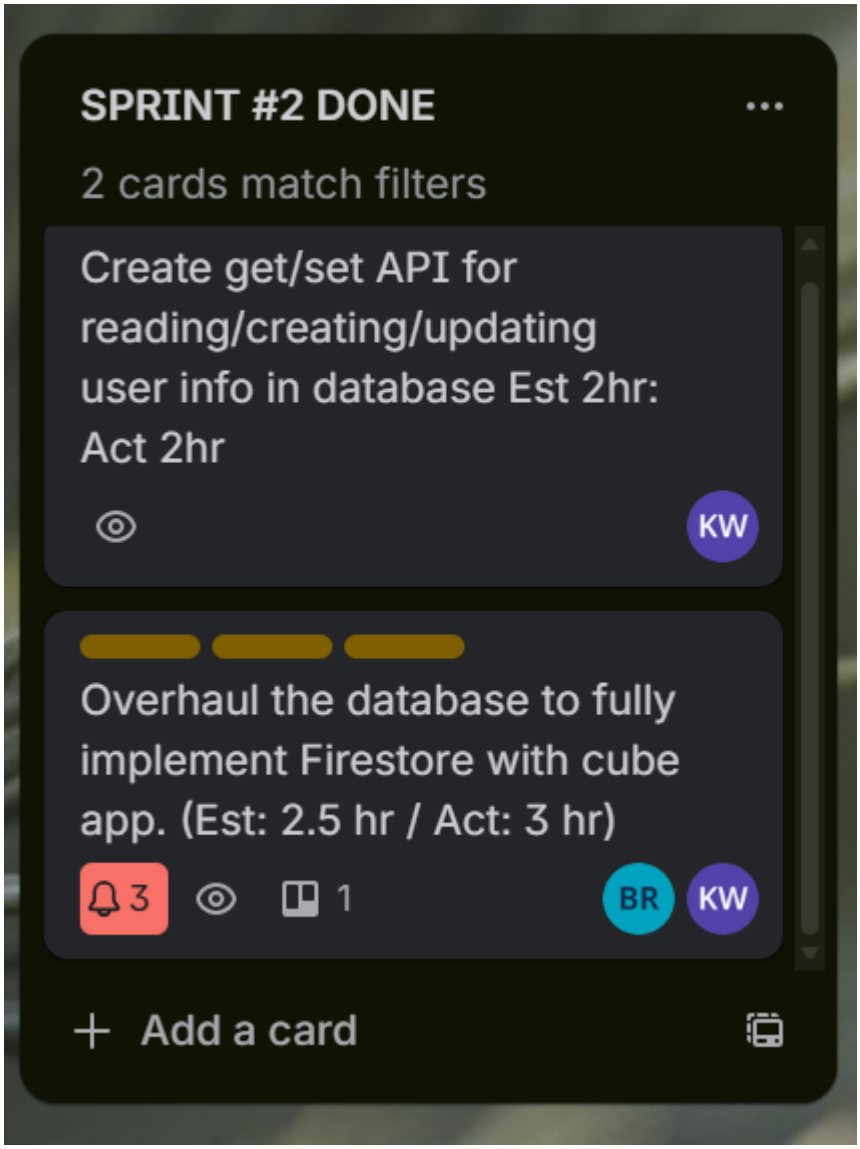
- **[Bryce Reid] Board Screenshot:**



- **[Perry Zhuo] Board Screenshot:**



- **[Kale Wyse] Board Screenshot:**

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## 2.2 Completed vs. Not Completed (Feature-Focused)

**Plan**

SPRINT 2:

- Web UI framework placeholders
- communication protocol between server + cube (or mock cube client)
- **DATABASE FRAMEWORK** - Ability to store necessary data on firebase securely
- **Cube-Client Control** - Ability to start, stop, & reset sessions and send session information to the server for storage
- **Web-Client Control** - Ability to select the current active task, create new tasks, and edit preset task times
- **USER STORIES FOR BASIC FUNCTIONALITY**

    - As a user i want the cube to send start, stop, and reset signals to the web app so that the app accurately tracks how long a cube has actively run a session.
    - As a user i want to the web app to send session time to the cube. So that session time can be easily changed.

- **END-TO-END FEATURES**

  *Cube Client Simulator*

    - Start Task
    - Stop Task
    - Reset Task

  *Web App*

    - Set Current Task
    - Create Preset Task
    - Update Current Task Time
    - Display Timer

  *Cube Timer Control*

**Completed in Sprint 2 (Feature)**

- ☐ **Web UI framework placeholders**
- ☐ **Cube-Client control** can trigger start stops and resets.
- ☐ **Web-Client control** exposes the endpoint with basic validation
- ☐ **Database framework** integration: data is written to the database
- ☐ **Server** can retrieve the stored data (GET from Firestore)
- ☐ **Authentication** firestore authentication for both client types as well as user login.

# 3. Technical Summary: What Was Implemented

This is a **short technical summary** of the **end-to-end feature** you built.

### Feature: Cube Client Simulator - Start Task

- **Collection:** N/A
- **What it does:** Triggers server to start timer for web app display ```

### Feature: Cube Client Simulator - Stop Task

- **Collection:** user_profiles -> uid -> session_history -> task_name & elapsed_time
- **What it does:** Triggers server to stop timer for web app display. Sends elapsed time recorded on the client to the server to store in database.

### Feature: Cube Client Simulator - Reset Task

- **Collection:** user_profiles -> uid -> current_task
- **What it does:** Triggers server to reset timer for web app display. Requests current task data from the server.

### Feature: Web App - Set Current Task

- **Collection:** user_profiles -> uid -> current_task
- **What it does:** Triggers server to update the current task in the database.

### Feature: Web App - Create Preset Task

- **Collection:** user_profiles -> uid -> presets -> task_name -> task_color & task_time
- **What it does:** Triggers server to create a new task preset with the input task name (task_time default to 10 min & task_color default to #ffaa00) in the database.

### Feature: Web App - Update Current Task Time

- **Collection:** user_profiles -> uid -> presets -> task_name -> task_time
- **What it does:** Triggers server to update the task time field in the current task preset in the database.

### Feature: Web App - Display Timer

- **Collection:** user_profiles -> uid -> presets -> task_name -> task_time
- **What it does:** Web app displays the task time associated with the current task. Display timer is controlled by CUBE client start, stop, & reset signals.

### Feature: [Cube Timer Control]

- **Collection:** N/A
- **What it does:** controls time on the hardware client

## Data Model (Firestore)

- **Document shape:**
  Example JSON that represents **one document** in the collection (or the schema you structured):

```
{
  "current_task": "Study",
  "presets": {
    "Flow State": {
      "task_color": "#ffaa00",
      "task_time": 2100
    },
    "Meditation": {
      "task_color": "#ffaa00",
      "task_time": 3355
    },
    "Relax": {
      "task_color": "#ffaa00",
      "task_time": 900
    },
    "Study": {
      "task_color": "#ffaa00",
      "task_time": 600
    }
```

```
    },
    "session_history": {
      "elapsed_time": 8,
      "task": "Flow State"
    },
    "user_info": {
      "email": "bryceroberttaylorreid@gmail.com",
      "first_name": "Bryce",
      "last_name": "Reid",
      "role": "user"
    }
  }
```

**Why this structure?** We used Firebase Authentication to validate users via email. Then with unique generated uuids we store the information we need. For basica usage we store preset tasks, connected cubes, current session, and session history.

- **Input (Client → Server):**
  Example JSON the CUBE client sends:

```
    {
      "action": "stop",
      "elapsed_seconds": self.elapsed_seconds
    }
```

  Example JSON the Web client sends:

```
    {
      "task_name": "chill time",
      "task_time": 600,
      "task_color": "#ffaa00"
    }
```

- **Output (Server → Client):**
  Example response the CUBE client receives after a successful create or read:

```
  {
    "message": "Meditation task reset",
    "task_name": "Meditation",
    "task_time": 3355
  }
```

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## 4. End-to-End Flow (What Was Demoed)

**Every JSON request from the cube client is accompanied by the cube UUID to link the session activity to a particular user profile.**

**This is a brief description of the Cube client interacting with the server to start a task session**

1. **Client** cube sends a reset request (e.g., POST `/feature`) with a valid **payload** (JSON).
2. **Server** responds with a json with the current time and current task information.
3. **Client** Displays current task via GUI.
4. **Client** Start button is pressed on client GUI which sends a start task request to the server and Client starts a timer to measure time elapsed along with displaying the task has begun via a faux LED in the GUI.
5. **Server** Once server receives a start task, the server starts a timer to measure elapsed time for web-app display and returns a json to the client that the task has started.
6. **Client** Once the stop button is pressed the client checks if a task is running (client does nothing if task has not been started) and sends a json stop request to the server which has the elapsed time within the json payload.
7. **Server** Server sends a json POST request to the database with the task_name and elapsed time which is saved in latest session database.

**This is a brief description of the Web-app client interacting with the server to change the current task**

1. **Client** On the web-app client the user selects a different task from the drop down menu this sends a POST with a valid **payload** (JSON) of the new task to update the current active task.
2. **Server** The server validates the uid for authentication.
3. **Server** The server updates the database current_task section.
4. **Server** The server updates the databases task_time section to correlate with the task selected.

### Bounded Read:

In Sprint 2, you were required to demonstrate a **bounded read** (e.g., `.limit()`, `.where()`, or pagination). Describe what you implemented:

**What we did:**

- Bound email data to 50 characters, task names to 30 characters : these were done to disable users from adding names of 'infinite' length to prevent attempting to store very large files.
- Bound task time to 99 minutes to prevent users from adding task times of 'infinte' time
- Bound task colour to values within the RGB hex values so users were not adding invalid colours to the cube itself.

### Communication between REST API on ESP32 with TEST FLASK SERVER

https://www.youtube.com/shorts/B0Q0dJrN8rY

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## 5. Sprint Retrospective: What We Learned

### 5.1 What Went Well

- **Item 1:** We were able to hit our targeted goals successfully in trello.

- **Item 2:** We managed merge conflicts reasonably well.
- **Item 3:** Merges were all handled by second parties.
- **Item 4:** Successfully managed communication mismatch.
- **Item 5:** Code Seperation handled effectively.
- **Item 6:** Wrapers used to manage complexity.
- **Item 7:** Cube look dope.

### 5.2 What Didn't Go Well

- **Item 1:** Planning was rushed, as a result documentation was a bit messy resulting in some communication errors
- **Item 2:** Towards the end we crammed to ensure feature completion.
- **Item 3:** Adding a feature after the inital sprint meeting resulted in addtion communcation mishaps.
- **Item 4:** We didn't come to clear enough agreements on DoDs for features.

### 5.3 Key Takeaways & Sprint 3 Actions

| Issue / Challenge | What We Learned | Action for Sprint 3 |
|---|---|---|
| **Rushing planning stage** | Results in potential misuse of time implementing features that may not be required | spend more time ensuring architecture and documentation are done well before beginning work |
| **Manage DoDs** | Unclear definitons of done results in miscomunication. | Assign *End User* and *Product Owner* roles to ensure clear vision of user requirements and clear vision of implementation strategies. |
| **Cramming** | Not clearly defining what tasks should are assigned led to delayed completion. | We should clarify roll requirements so that each member has a clear idea of what is being implemented. |
| **Feature Creep** | Not managing mid-sprint feature creep led to unnecessary stress. | In the future we should have a meeting to ensure proper handling of feature creep. |

## 6. Sprint 3 Preview

Based on what we accomplished (and what we didn't), here are the **next Sprint 3 priorities**:

- **Hardware Integration**: Ensure working communication between SERVER and hardware CUBE client. Color and pattern should be controlable from web UI.
- **End to End Cube Management**: Address ability to select and diffferentiate cubes. There shouldn't be error when multiple users connect to same cube.
- **UI Mangement**: UI needs to connect to several more routes. Delete task.
- **Finish PCB**: Additional.